

APURB AGARWAL

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🌐 apurb-agarwal

🌐 agarwalapur

EDUCATION

Year	Academic Qualification	Institute	CPI/%
2024	B.Tech, Electrical Engineering	Indian Institute of Technology Kanpur	9.24/10.0
2020	Senior Secondary(CBSE)	LBS School, RK Puram, New Delhi	97.2%
2018	Secondary (ICSE)	St. Michael's School, Siliguri	98.6%

ACHIEVEMENTS

- Received **Academic Excellence Award** in **Aug'21** and **May'22**, awarded by **IIT Kanpur**.
- Won **Silver Medal** in **Inter IIT Tech-Meet 10.0** organised by IIT Kharagpur in **2022**.
- **All India Rank 1127** in **Joint Entrance Examination (Advanced) 2020** among 250,000 candidates.
- **All India Rank 1088** in **Joint Entrance Examination (Main) 2020** among 1.1 million candidates.
- **Kishore Vaigyanik Protsahan Yojana** fellowship **2018** awarded by DST, Government of India.
- Secured **State Rank 3** in **Indian Certificate of Secondary Education(ICSE) 2018**.
- Achieved **Global Rank 2364** in **Google Kickstart Coding Challenge Round C 2022**.

INTERNSHIP

Research and Development Intern — Eigenlytics

May 2022 – Present

- Implemented **skew-correction** of scanned documents after identifying text regions using **Craft detection**.
- Used **Edge Detection** and **Hough Transform** techniques to achieve an accuracy of over **98%**.
- Worked on **Document Classification** using **TFIDF** for identifying the document from **5** different categories.

PROJECTS

Silicon Labs' Social Entrepreneurship Challenge

Feb 2022 – March 2022

Inter IIT Tech-Meet 10.0

IIT Kharagpur

- Worked in a team of **10** to provide an end-to-end **Internet of Things** based solution for a **social cause**.
- Implemented a hardware **prototype** for **real-time monitoring** of **6** vitals of patients in hospitals.
- Proposed an **Early Warning System** based on a **risk score** in case the patient's condition deteriorates.
- Devised the complete **business plan** and **market strategy** to launch the product in the market.

Automatic Music Transcription using Deep Learning ↗

June 2021 – July 2021

Summer Project — Electronics Club

IIT Kanpur

- Implemented **Music Transcription** on **MAPS** dataset after preprocessing it using various **transforms**.
- Explored various neural network architectures like **CNN**, **RNN** and **LSTM** using **Tensorflow** framework.
- Compared the results obtained by the various models on **accuracy**, **recall**, **precision** and **f-score**.

Classification of IMDb Reviews ↗

July 2021 – Aug 2021

- Vectorized text corpus using **Tokenizer** and used **Embedding** and **Global Average Pooling**. *Self Project*
- Trained the model on **40000** reviews and achieved a **validation accuracy** of about **89%**.

Maze Solver for Micromouse Robot ↗

Nov 2021 – Dec 2021

- Explored an **algorithm** for finding path in a previously **unknown maze** environment. *Self Project*
- Implemented **Floodfill Algorithm** from scratch in **C++** based on **Breadth First Search**.

COURSEWORK

*:Coursera **:Upcoming

Data Structures & Algorithms	Algorithms-II**	Supervised Machine Learning*
Fundamentals of Computing	Real Analysis	Advanced Learning Algorithms*
Probability & Statistics	Linear Algebra	Neural Networks & Deep Learning*

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Julia

Utilities and Frameworks: Tensorflow, numpy, matplotlib, pandas, scikit-learn, Git, PyTorch

EXTRACURRICULAR ACTIVITIES

- Senior Member at Team ViSiON, which focuses on developing autonomous ground vehicles.
- Secretary at Electronics Club, responsible for conducting various events for campus community.
- Participated and won in various inter-school quizzes at national level.